04 / Grounding

What connects the representation or interpretation to physical evidence?

EX02 / UMI | Source analysis | C1 | P: paper; W: project; R: repository

Prediction / representation	Matched reference	Difference / ambiguity	Conclusion / scope limit

1. Comparison boundary

Name the prediction/measurement and reference. Match conditions; identify simulated versus physical evidence.

Same trained tossing policy; latency matching enabled versus delays set to zero. Physical object outcomes are the reference, not plausible reconstructed motion. [P V-B]

2. Quantities and validity

Declare quantity, units, frame, tolerance and measurement limits. Distinguish a chosen tolerance from observed uncertainty.

Use object success fraction and delay in seconds; retain arm/gripper release offsets. Manual outcome judgments have limits; the bin rule is in App A-C. [P Fig 5]

3. Results and uncertainty

Give paired outcomes, trial counts and uncertainty or limits. If untested, record the protocol and missing evidence explicitly.

105/120 matched versus 69/120 unmatched: +30 percentage points. Counts are object outcomes; episode dependence and run-to-run uncertainty are not resolved by this fraction. [P V-B]

4. Interpretation and challenge

Bound the conclusion and consider another cause of the difference. Choose a diagnostic test or scope restriction.

Timing supports this comparison, not universal transfer. SLAM error, release timing and scene changes remain alternatives; inspect one miss rather than infer reliability from throughput. [P V-B; R]

Review: Could sensing error, reset differences or selection explain the discrepancy instead of model error?

Example response: A 30-point difference concerns this timing ablation; objects in one episode need not be independent.

Worksheet 4 / Grounding – evidence and team review

EX02 / UMI | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX02 Project: UMI Claim ID: C1 Revision: draft 2

Mode: Source analysis Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: selected paper text/tables checked 20 Oct 2026; no raw dataset download or independent experiment.

Evidence ID	Artifact/source + location (clickable)	What it supports; limitation / access gap
P	arxiv.org/html/2402.10329v3	Paper: method/results; reported evidence, not our measurement.
W	real.stanford.edu/umi	Project: system/videos; selected examples do not establish reliability.
R	github.com/real-stanford/universal_manipulation_interface	Repository: linked route; code, raw data and current commit not audited.

Next action / owner / verification:

Analyst: verify 105/120 and 69/120 under the same trained policy; retain the manual outcome and retry rule.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

2. Name / student ID: _____

Signature: _____ Date: _____

3. Name / student ID: _____

Signature: _____ Date: _____

4. Name / student ID: _____

Signature: _____ Date: _____

5. Name / student ID: _____

Signature: _____ Date: _____

6. Name / student ID: _____

Signature: _____ Date: _____

7. Name / student ID: _____

Signature: _____ Date: _____

8. Name / student ID: _____

Signature: _____ Date: _____

P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.